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Video Piracy Websites Detection using Continual Learning with Elastic Weight Consolidation

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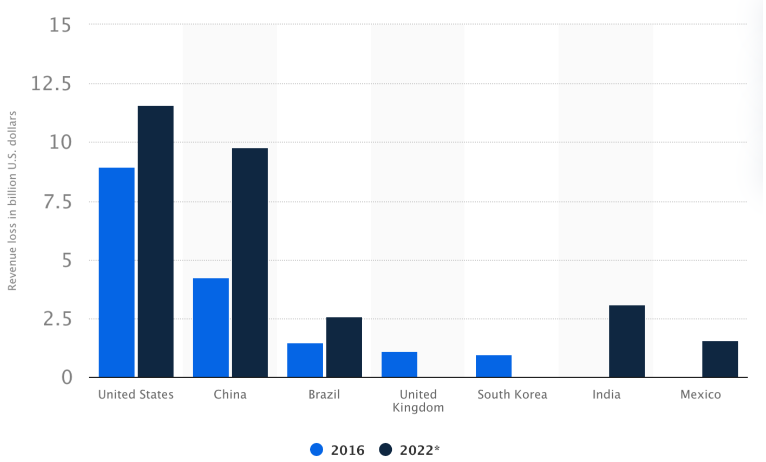
1.1 Video Piracy Websites (VPWs)



- VPWs distribute video content without permission from copyright holders or legitimate purchasers.
- The U.S. economy loses at least USD **$29.2 billion** in revenue each year due to pirated material [Blackburn et al. 2019].
- China is expected to lose USD **$9.78 billion** through piracy in the movie and online TV revenue in 2022 [Statista 2023].

http://www.digital-digest.com/blog/DVDGuy/category/movies/high_definition_dvd/
https://www.statista.com/statistics/778342/global-online-tv-movie-revenue-loss-piracy-country/

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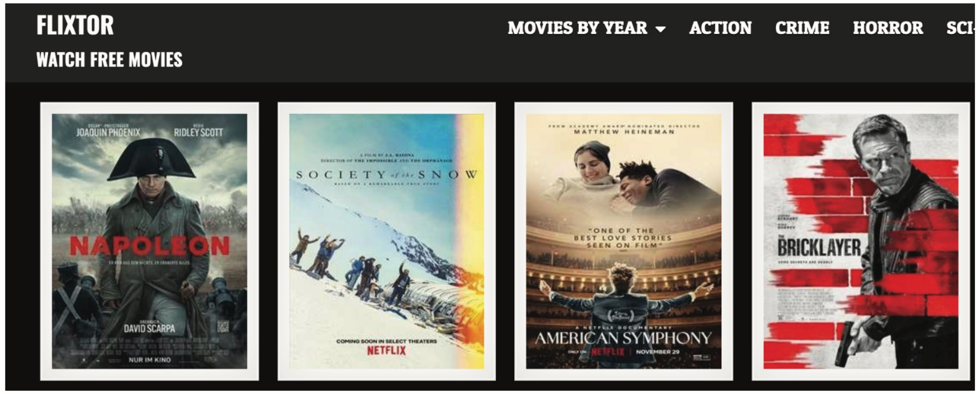
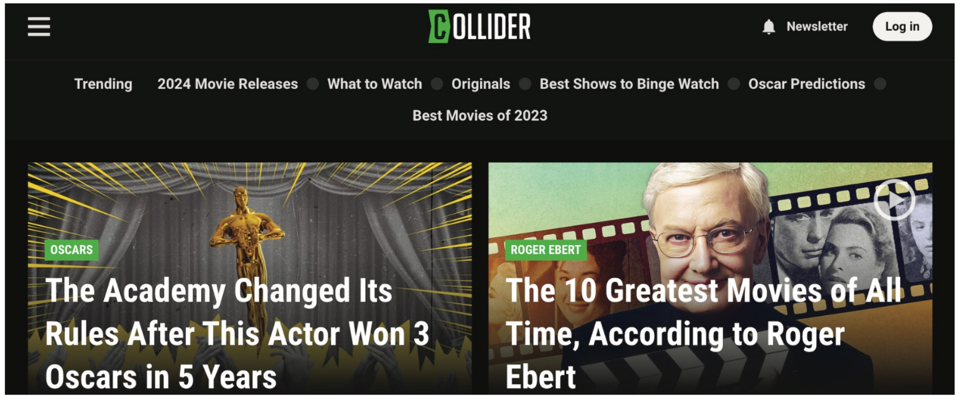
1.2 Problem

- **Problem:** Given features of a website, detect whether it is a VPW
- **Input:** Website features including website content text, website's domain, title, keyword, and URL
- **Output:** Binary classification of whether a website is a VPW or not



VPWs



Other Websites Found by Searching Keywords

1.3 Challenges

i. Performance drop with single training

- If we train a model only once, the performance of the model may decrease **when new datasets of VPWs arrive**.

ii. Large runtime and resource cost

- Training a new model on both new and old data together whenever new datasets are collected would be highly time-consuming and resource-inefficient.

iii. Lack of old data accessibility

- Some or all old training data may be lost or unavailable due to storage limits or other factors.

1.4 VPW Detection

- We design a **Video Piracy Websites Detection (VPWD)** architecture that supports the end-to-end process of possible VPW collection to detection.
- We build a dataset of VPWs and non-VPWs by crawling web pages from search engine results using video piracy-related keywords.
- We incorporate **continual learning** methods into VPW detection to address the three challenges above.

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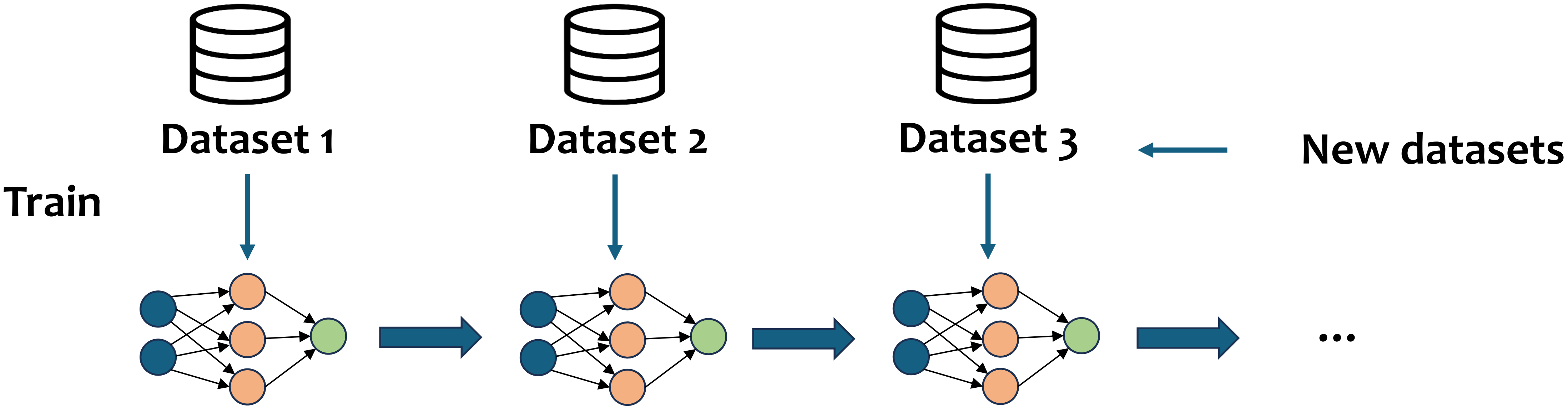
1.4 VPW Detection



- Continual learning
  - Train sequentially on multiple tasks (or datasets).
  - Learn from new tasks without explicitly retraining on previous tasks.
  - Achieve good performance on both new tasks and old ones.

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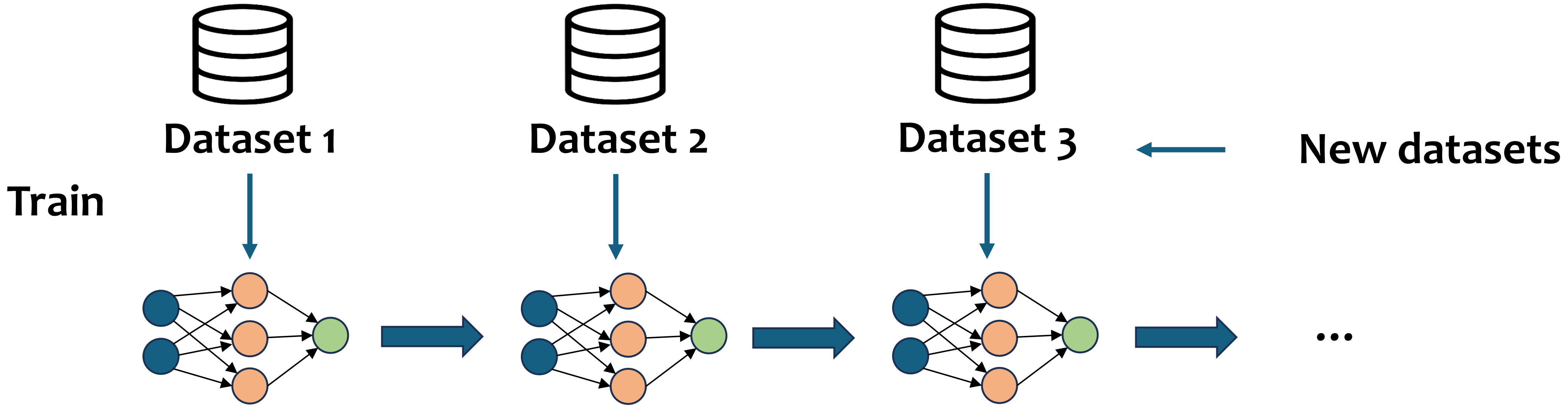
1.4 VPW Detection



- Address the challenges by:
  i. training a model continually
  ii. doing so time- and resource-efficiently by training only on new data
  iii. not training on old data

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1.5 Contributions

- We design the **first continual learning-based Video Piracy Websites Detection (VPWD)** architecture.
- We build the **largest** real-world dataset of features targeting VPWs, available at https://github.com/LucyDYu/VPWD.
- We conduct experiments on our dataset using continual learning algorithms.

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2 Related Work

- Website detection
  - **Detection purposes**
    - Piracy website detection
    - Phishing website detection
    - Malicious website detection
  - **Detection methods**
    - Non-machine learning approaches [S.-K. Choi et al. 2020; Kim et al. 2021]
    - Traditional machine learning approaches [Kumar et al. 2020; Shabudin et al. 2020]
    - Deep learning approaches [S. Zhang et al. 2022; C. Wang et al. 2022; Zhao Li et al. 2022; Ejaz et al. 2023]
- **Continual learning (CL)**
  - Regularization-based methods [Kirkpatrick et al. 2017; Huszár 2018]
  - Architecture-based methods [Rusu et al. 2016; Yoon et al. 2020]
  - Replay-based methods [Rebuffi et al. 2017; Lopez-Paz et al. 2017]

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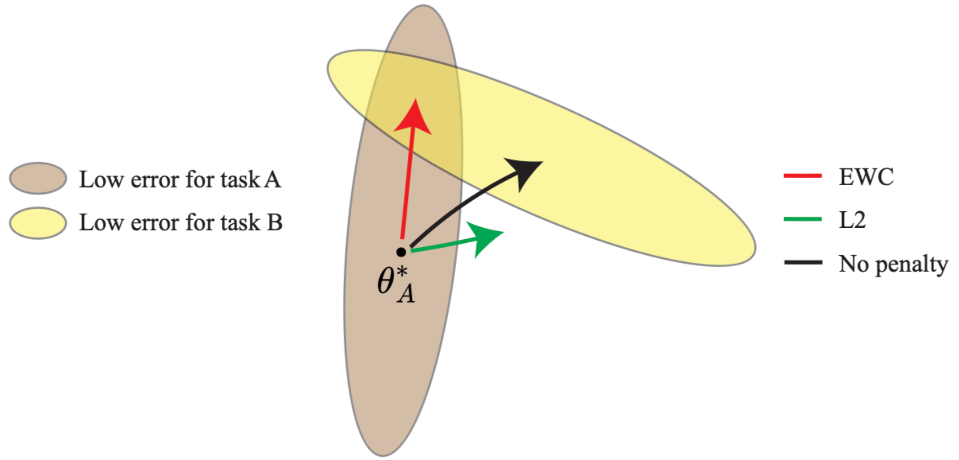
2.1 Regularization Based – EWC

- Impose additional constraints on the parameters through regularization terms, adding extra costs associated with parameter changes.
- Elastic Weight Consolidation (EWC) [Kirkpatrick et al. 2017]: After training on task A and obtaining optimal parameters $\boldsymbol{\theta}_A^*$, task B is trained with the following loss:
- $\mathcal{L}(\boldsymbol{\theta}) = \mathcal{L}_B(\boldsymbol{\theta}) + \sum_i \frac{\lambda}{2} F_{A,i}\left(\theta_i - \theta_{A,i}^*\right)^2$



Graphical illustration of EWC. Figure adapted and redrawn based on [Kirkpatrick et al. 2017].

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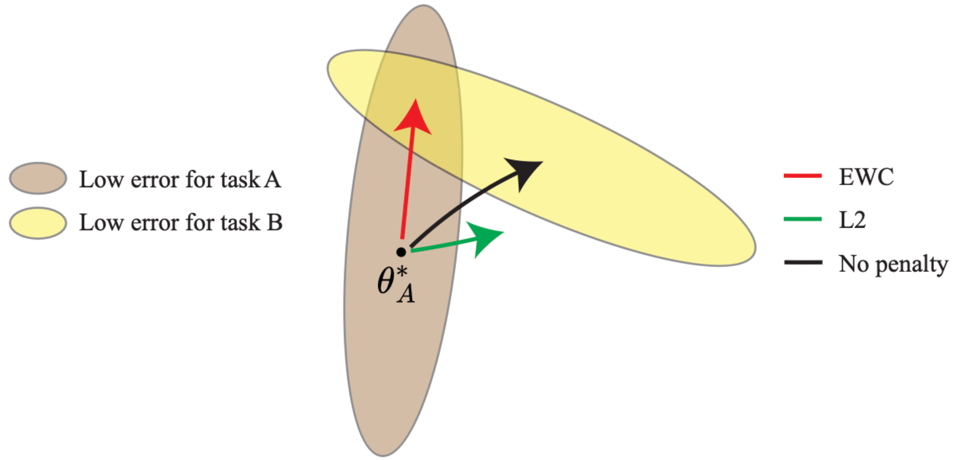
2.1 Regularization Based – EWC

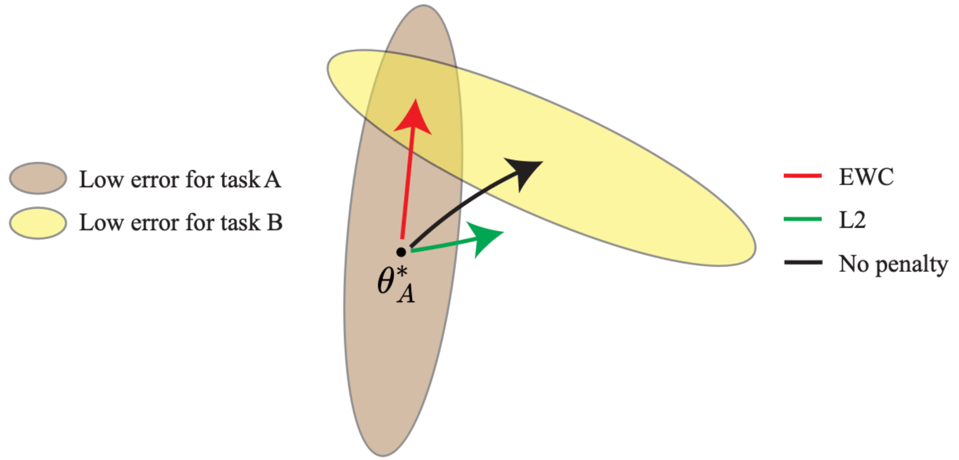
- $\mathcal{L}(\boldsymbol{\theta}) = \mathcal{L}_B(\boldsymbol{\theta}) + \sum_i \frac{\lambda}{2} F_{A,i}\left(\theta_i - \theta_{A,i}^*\right)^2$
- $F_{A,i}$ is the diagonal of the Fisher information matrix, obtained by sampling from the dataset of task A and calculating the expectation of squared gradient of log-likelihood of each parameter.
- $F_{A,i}$ reflects the importance of parameters to the first task A.



Graphical illustration of EWC. Figure adapted and redrawn based on [Kirkpatrick et al. 2017].

2.2 Regularization Based – L2

- L2 penalty: can be considered to be a special case of EWC and imposes equal constraints on all parameters [Lee et al. 2017]:
- $\mathcal{L}(\boldsymbol{\theta}) = \mathcal{L}_B(\boldsymbol{\theta}) + \sum_i \frac{\lambda}{2}\left(\theta_i - \theta_{A,i}^*\right)^2$



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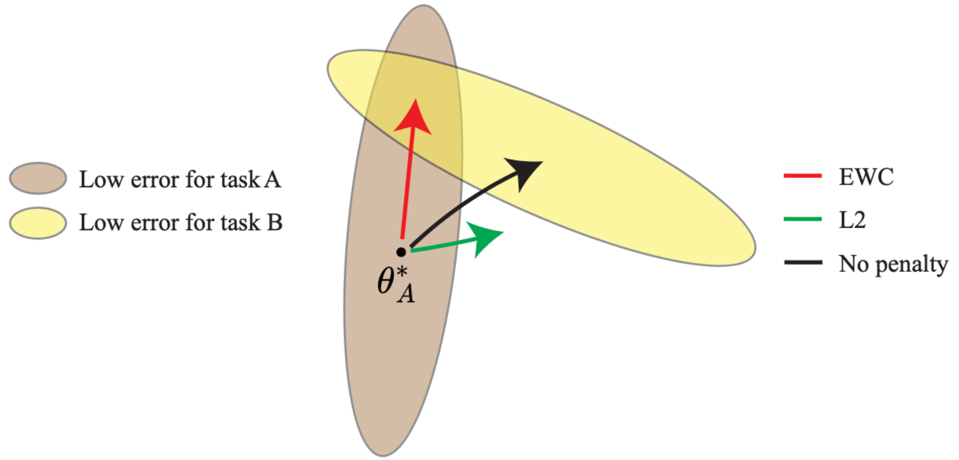
2.3 Regularization Based – EWC^{H}

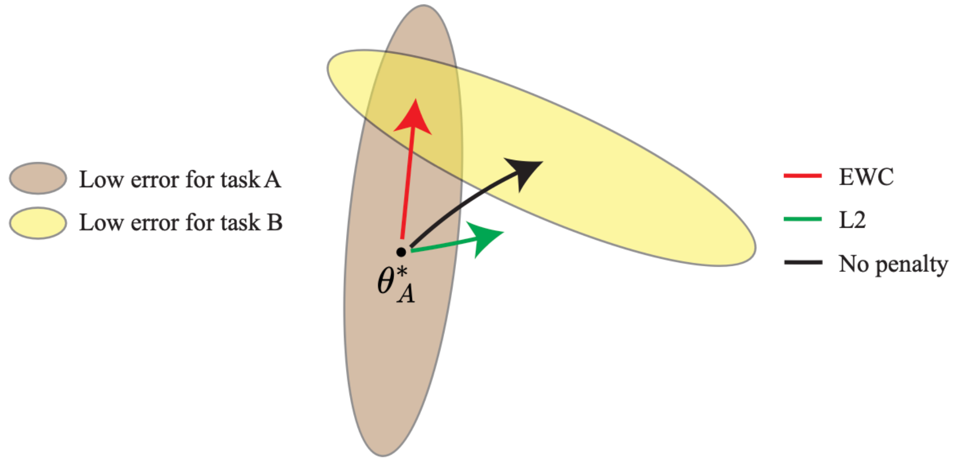
- EWC loss on three tasks:
- $\mathcal{L}(\boldsymbol{\theta}) = \mathcal{L}_C(\boldsymbol{\theta}) + \sum_i \frac{\lambda}{2} F_{A,i}\left(\theta_i - \theta^*_{A,i}\right)^2 + \sum_i \frac{\lambda}{2} F_{B,i}\left(\theta_i - \theta^*_{A,B,i}\right)^2$
- **Limitation**: The model is restricted from moving away from **multiple optimal points** as the number of tasks increases
- **Performance Issue**: This limitation leads to suboptimal performance [Huszár 2018]
- **Sensitivity**: The model is sensitive to task order [Huszár 2018]



Graphical illustration of EWC. Figure adapted and redrawn based on [Kirkpatrick et al. 2017].

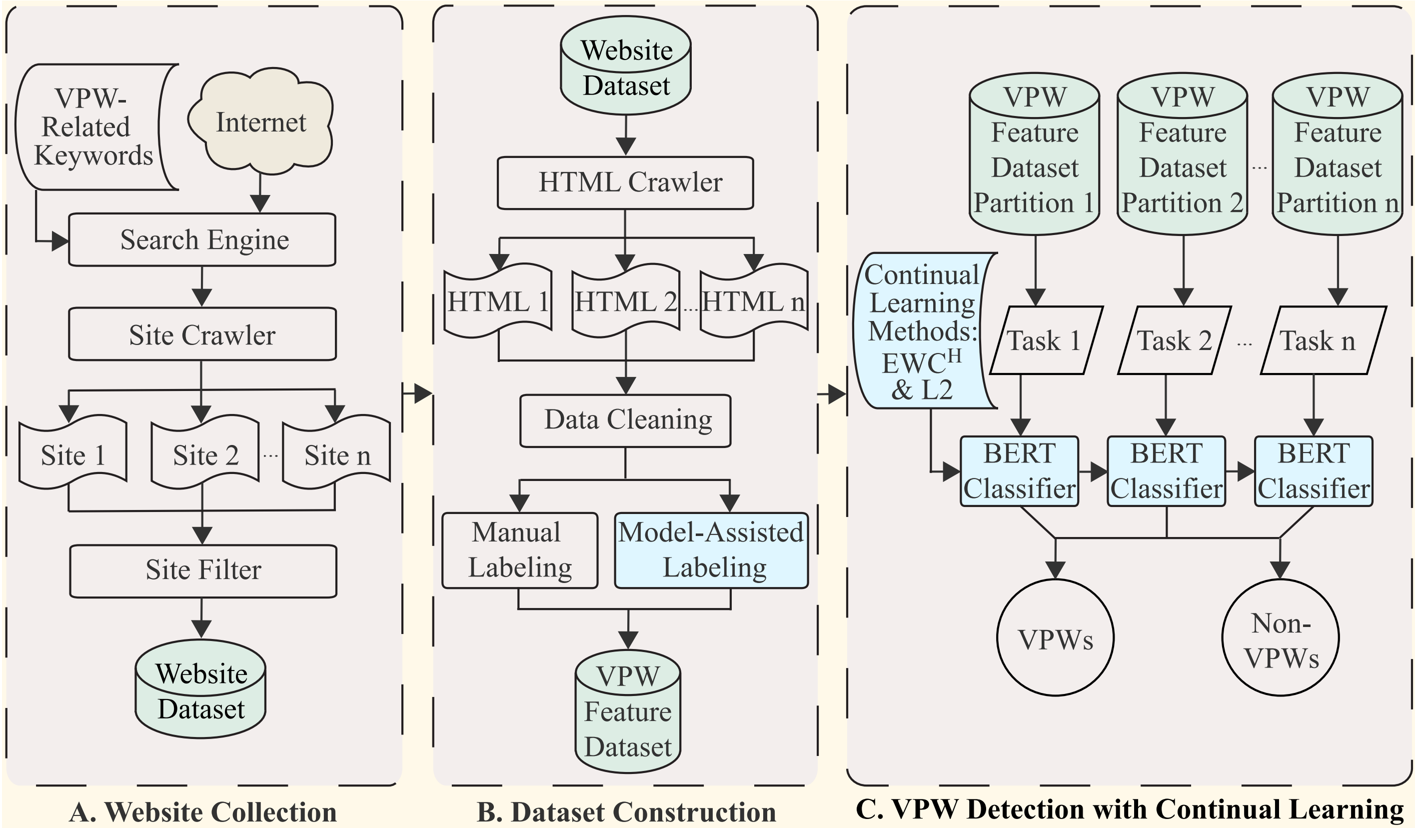
2.3 Regularization Based – EWC^H

- EWC^H: An improved version of EWC by [Huszár 2018].
- Loss: $\mathcal{L}(\boldsymbol{\theta}) = \mathcal{L}_C(\boldsymbol{\theta}) + \sum_i \frac{\lambda}{2}(F_{A,i}+F_{B,i})\left(\theta_i - \theta^*_{A,B,i}\right)^2$
- EWC^H considers **only the previous optimal point** and maintains only a single regularization term, improving calculation efficiency and demonstrating better performance.



Graphical illustration of EWC. Figure adapted and redrawn based on [Kirkpatrick et al. 2017].

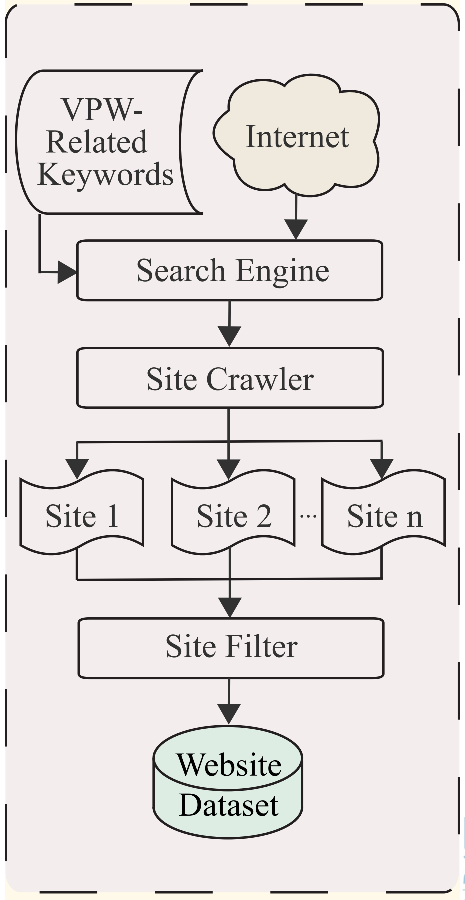
3 Methodology



Video Piracy Website Detection (VPWD) architecture

3 Methodology

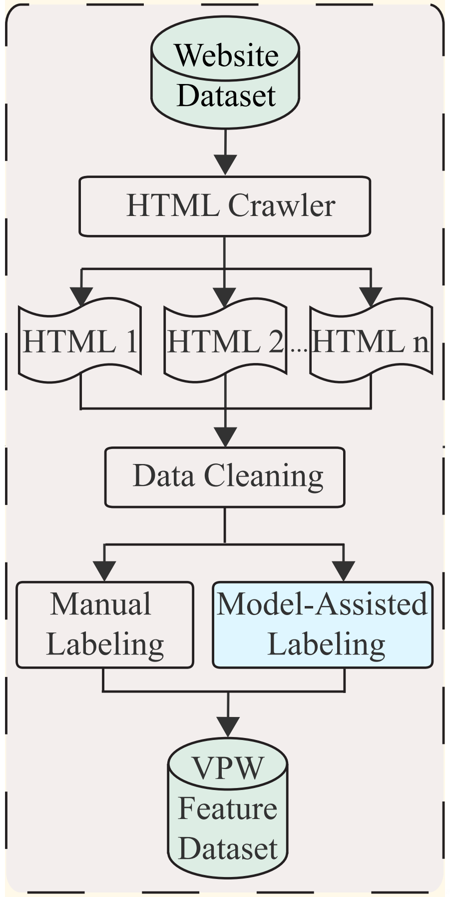
- We use a site crawler to collect URLs from search engine results using VPW-related keywords and get 103,376 URLs.
- These keywords include both English and Chinese phrases, such as “free movie” and “免费观看” (watch for free).
- Remove duplicate URLs based on domain names to keep one URL per domain.
- A dataset of 14,254 potential VPWs.



A. Website Collection

3 Methodology

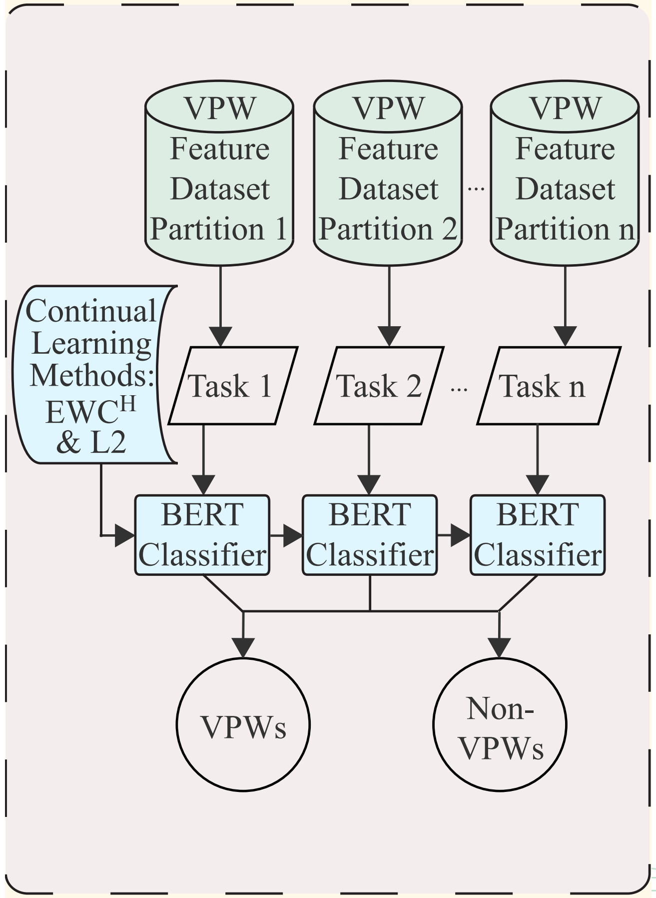
- Use an HTML crawler to collect HTML files from the URLs
- **Manual labeling**: 254 VPWs and 313 non-VPWs
- **Model-assisted labeling**: in a classical semi-supervised learning fashion [Yang et al. 2023]
  - Fine-tune a BERT classifier [Devlin et al. 2019] on the manually labeled data to learn relevant VPW features
  - Infer labels for the unlabeled data
  - Only select labels from the model with a probability over 98%
- Obtain 5,759 labeled websites with 2,794 VPWs and 2,965 non-VPWs



B. Dataset Construction

3 Methodology

- Partition our dataset into subsets
- Fine-tune a BERT classifier sequentially on them, using CL methods like EWC^H [Huszár 2018]
- As more keywords are added, our VPWD architecture can
  - crawl more data
  - label them (manual labeling is optional for improving labeling quality)
  - continually learn VPW features to detect them



C. VPW Detection with Continual Learning

4 Experiments – Dataset

- 52 VPW-related keywords
- Partition the dataset by keywords: KeywordA, KeywordB, and KeywordC
- Partition the dataset by languages: LanguageA (English) and LanguageB (Chinese)

Table 2. Dataset Distribution - Keywords and Languages

Dimension		VPWs	Non-VPWs	Total	VPW Rate
Keywords	免费观看 (watch for free)	136	23	159	85.5%
	free movie	49	47	96	51.0%
	online series	57	76	133	42.9%
	Other keywords	2,552	2,819	5,371	47.5%
Languages	English	518	2,144	2,662	19.5%
	Chinese	2276	821	3,097	73.5%
Total		2,794	2,965	5,759	48.5%

4 Experiments – RQ1

- RQ1. How does a fixed model perform on new datasets?
- When the model is trained on one dataset, the performance on this dataset is high, but the performance on other datasets is much lower.

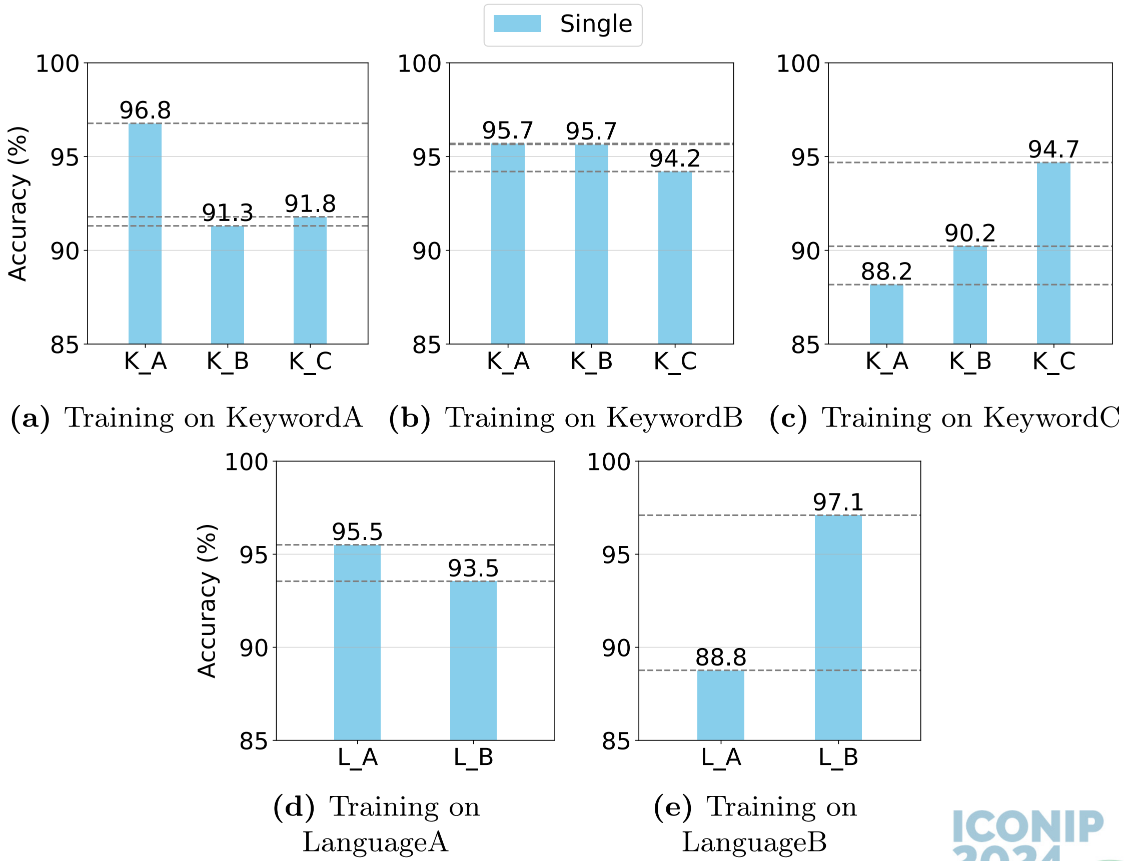


Fig. 4. Test accuracy when training on one dataset.

4 Experiments – RQ1

- Fig. 4a shows that the model trained on KeywordA achieves 96.8% test accuracy on KeywordA, while those of KeywordB and KeywordC are 91.3% and 91.8%.
- Fig. 4e shows that the model trained on LanguageB achieves 97.1% test accuracy on LanguageB, while that of LanguageA is 88.8%, **dropping over 8%**.

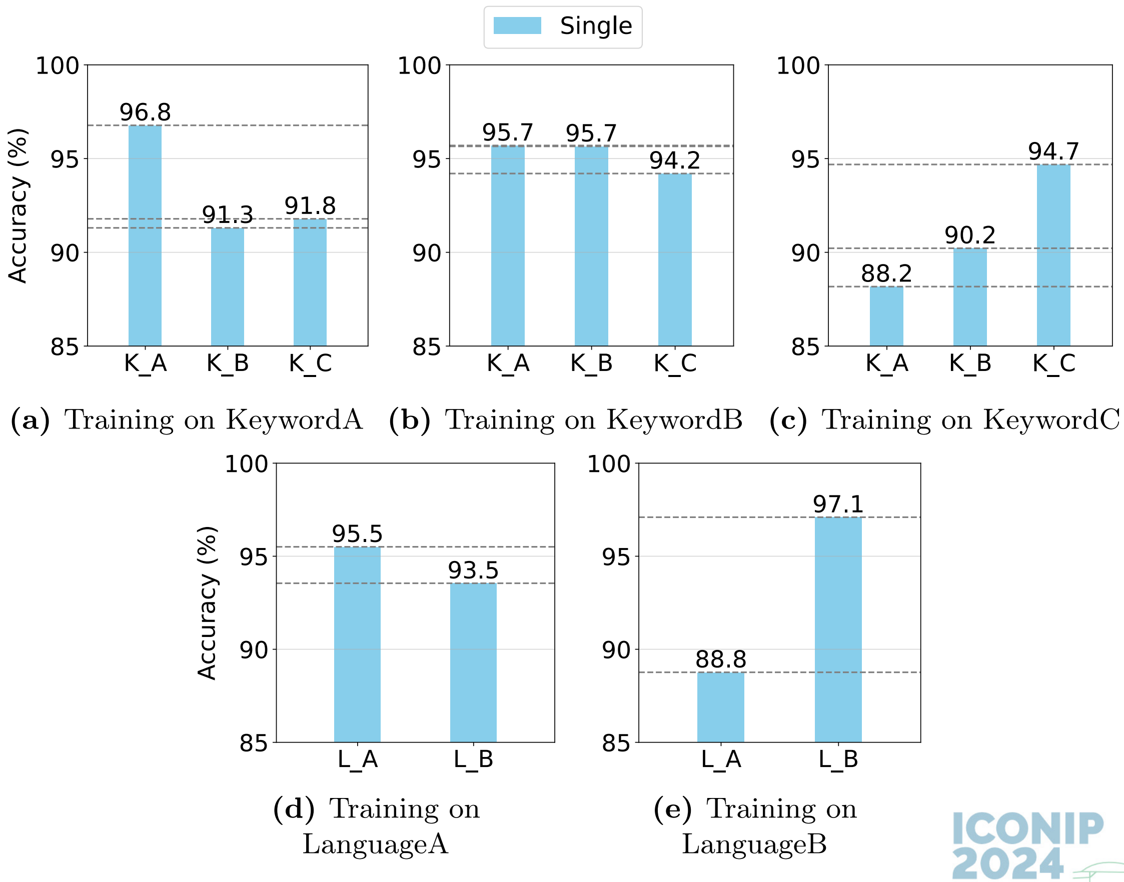


Fig. 4. Test accuracy when training on one dataset.

4 Experiments – RQ2

- RQ2. What is the effect of the EWC^{H} method when the model is continually trained on multiple datasets?
- With EWC^{H}, when the model continually trains on a new task, performance on old tasks remains high.

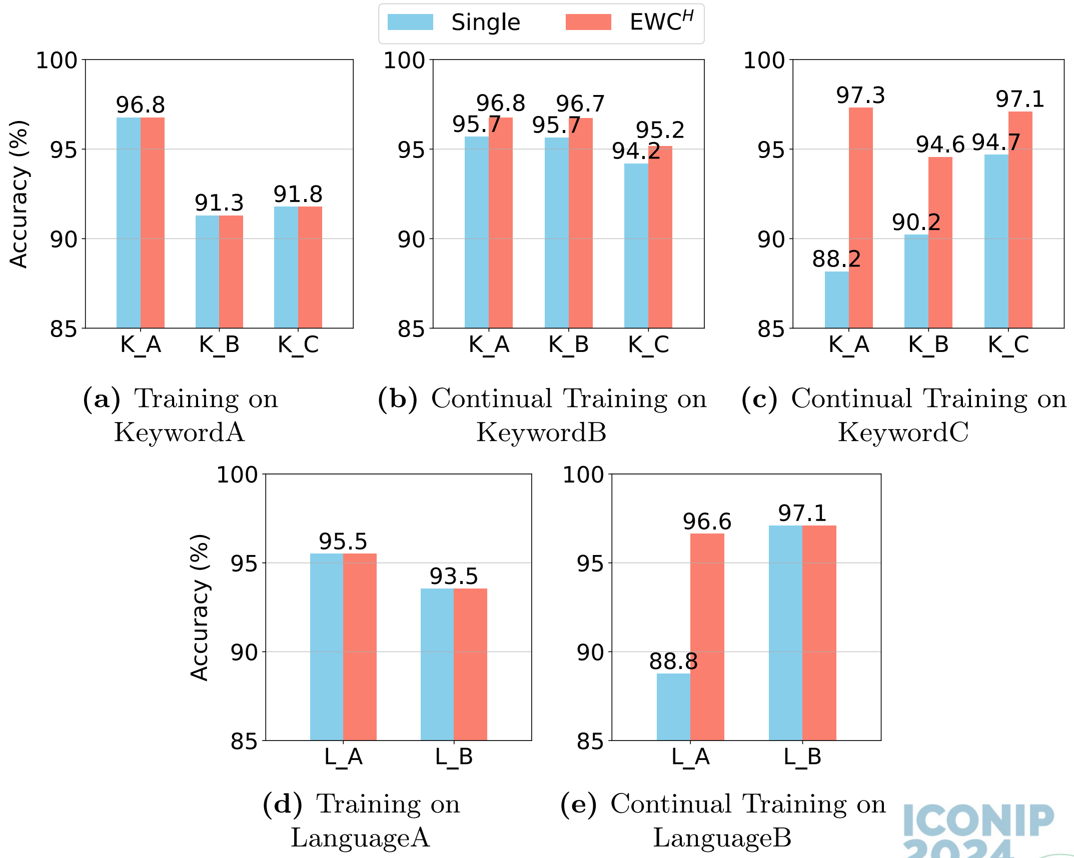
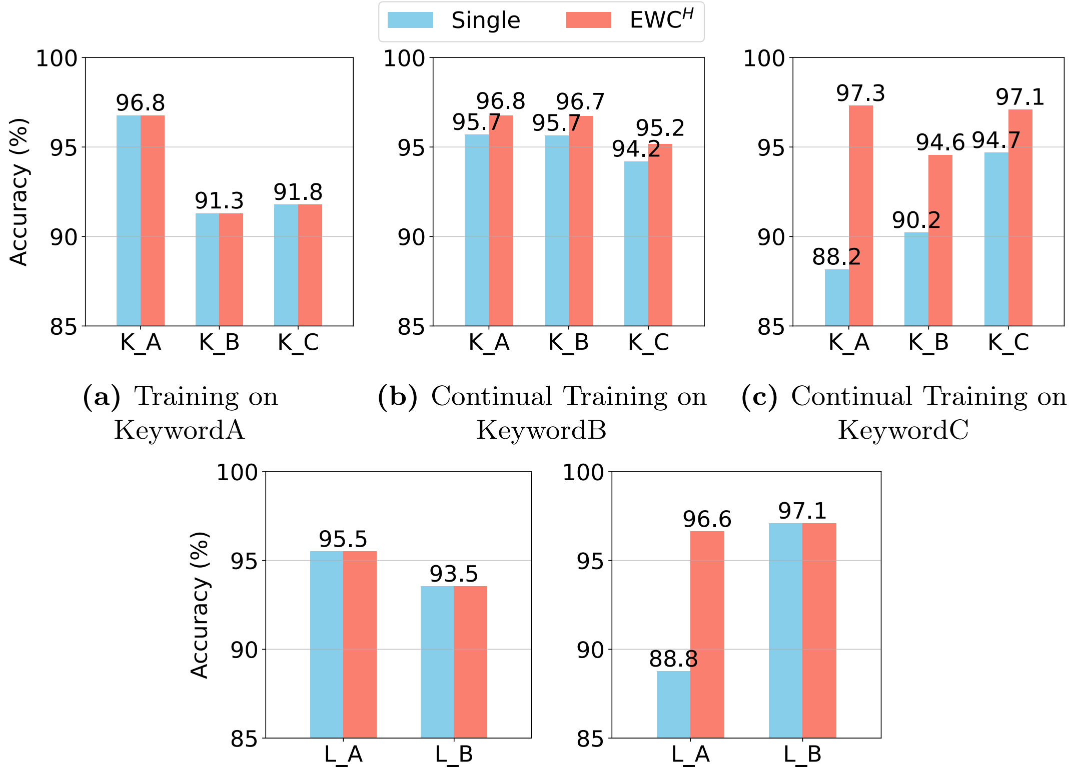


Fig. 5. Test accuracy when training with EWC^{H}.

4 Experiments – RQ2

- Fig. 5c shows that the model trained on KeywordC achieves 97.1% test accuracy on KeywordC, and 97.3% and 94.6% on KeywordA and KeywordB.
- Fig. 5e shows that the model trained on LanguageB achieves 97.1% test accuracy on LanguageB, and 96.6% on LanguageA.
- **This result demonstrates the effectiveness of EWC^H and the necessity of CL.**



(a) Training on KeywordA
(b) Continual Training on KeywordB
(c) Continual Training on KeywordC
(d) Training on LanguageA
(e) Continual Training on LanguageB

Fig. 5. Test accuracy when training with EWC^H.

4 Experiments – RQ3

- RQ3. How does EWC^H as a CL algorithm compare with other CL algorithms (L2) and non-CL methods (Fine-tuning)?
- EWC^H achieves the highest average test accuracy in both keyword partition and language partition experiments.
- EWC^H also achieves the highest average AUC value.

Table 3. Performance comparison of different methods on each task using keyword-based dataset partition.

Method	Test Accuracy				AUC Value				Precision	Recall	F1 Score
	KeywordA	KeywordB	KeywordC	Average	KeywordA	KeywordB	KeywordC	Average	Average	Average	Average
Fine-tuning	$97.1_{\pm.3}$	$94.7_{\pm.3}$	$96.0_{\pm.7}$	$95.9_{\pm.3}$	$96.2_{\pm1.1}$	$97.5_{\pm.2}$	$99.3_{\pm.1}$	$97.7_{\pm.4}$	$97.5_{\pm.7}$	$91.8_{\pm.1}$	$94.5_{\pm.4}$
L2	$95.7_{\pm.0}$	$92.9_{\pm1.4}$	$93.1_{\pm.3}$	$93.9_{\pm.5}$	$95.3_{\pm2.4}$	$97.3_{\pm.8}$	$97.6_{\pm.1}$	$96.7_{\pm.9}$	$93.3_{\pm.8}$	$90.9_{\pm.4}$	$92.1_{\pm.5}$
EWC^H	$\mathbf{97.5}_{\pm.3}$	$\mathbf{94.9}_{\pm.6}$	$\mathbf{96.9}_{\pm.7}$	$\mathbf{96.4}_{\pm.3}$	$\mathbf{98.0}_{\pm.8}$	$\mathbf{98.2}_{\pm.1}$	$\mathbf{99.4}_{\pm.1}$	$\mathbf{98.5}_{\pm.3}$	$\mathbf{97.8}_{\pm.4}$	$\mathbf{92.7}_{\pm.4}$	$\mathbf{95.1}_{\pm.3}$

Table 4. Performance comparison of different methods on each task using language-based dataset partition.

Method	Test Accuracy			AUC Value			Precision	Recall	F1 Score
	LanguageA	LanguageB	Average	LanguageA	LanguageB	Average	Average	Average	Average
Fine-tuning	$93.8_{\pm1.6}$	$\mathbf{97.5}_{\pm.4}$	$95.6_{\pm.6}$	$96.0_{\pm1.0}$	$\mathbf{99.3}_{\pm.0}$	$97.6_{\pm.6}$	$96.0_{\pm.7}$	$87.3_{\pm4.3}$	$90.9_{\pm2.3}$
L2	$\mathbf{95.3}_{\pm.4}$	$93.3_{\pm1.2}$	$94.3_{\pm.8}$	$\mathbf{97.2}_{\pm.2}$	$95.4_{\pm.7}$	$96.3_{\pm.4}$	$93.8_{\pm.9}$	$\mathbf{90.4}_{\pm.8}$	$91.9_{\pm1.0}$
EWC^H	$95.1_{\pm1.4}$	$96.8_{\pm.6}$	$\mathbf{96.0}_{\pm.9}$	$96.5_{\pm.6}$	$\mathbf{99.3}_{\pm.1}$	$\mathbf{97.9}_{\pm.3}$	$\mathbf{96.6}_{\pm1.2}$	$89.5_{\pm3.1}$	$\mathbf{92.6}_{\pm2.1}$

5 Discussions

- Complexity Analysis
  - EWC^H uses the Fisher information matrix.
  - The calculation is in $O(P \cdot T)$, where P represents the number of parameters and T represents the number of tasks.
  - Our VPWD architecture with continual learning capability is as efficient as without EWC^H.
- Limitations
  - EWC and its variants only use the diagonal of the Fisher information matrix and assume each parameter is independent of others [Lee et al. 2017]. This assumption is fragile and not true in general.
  - Require additional storage and memory to store the diagonal of the Fisher information matrix and the previous optimal parameters $\boldsymbol{\theta}^*$.

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6 Conclusion and Future Work

- Conclusion
  - Propose a CL-based VPWD architecture
  - Build a real-world dataset of features targeting VPWs
  - Conduct empirical experiments on our dataset
- Future Work
  - Propose new CL methods and employ existing ones
  - Collect data for other website detection tasks and let the model continue learning related features
  - VPWD ⟶ general-purpose website detection architecture

Thank you!

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